Einstein's General Theory of Relativity

Lecture 10: Einstein's Equation, Waves, and the Cosmological Constant

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Chapter 1

Lecture 10: Einstein's Equation, Waves, and the Cosmological Constant

Overview. A point mass in $2+1$ dimensions makes gravity geometrically visible: space is a cone whose missing angle records the mass. From that warm-up we ask why three spatial dimensions are special, reconstruct the conserved geometric side of Einstein's equation, and use the equivalence principle as a practical rule for carrying a scalar wave equation into curved spacetime. The wave must then gravitate in return, so we derive its stress-energy tensor and learn why pressure and shear belong among the sources of gravity. The closing questions return to the cosmological constant, Einstein's unstable static universe, light as a source, photon orbits, and the difference between spatial flatness and flat spacetime.

We use the signature

$$g_{\mu\nu} \longrightarrow \eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1) \tag{1.1}$$

in a local inertial frame, and we usually set $c = 1$. Whenever a factor of c clarifies a physical comparison, it will be restored.

1.1 A Point Mass Makes a Cone

Begin with empty flat spacetime, and for the moment look only at a spatial slice. Put one point particle into an otherwise empty $2+1$-dimensional universe. The source is concentrated at one point, so the natural question is immediate: what surface is flat everywhere except at a single point?

Question from the audience. What geometry is flat everywhere except at the position of the particle? [a]

Response. A cone. Cut a wedge from a flat sheet and identify the exposed edges. Every sufficiently small patch away from the tip is Euclidean, yet a loop surrounding the tip detects a missing angle. In $2+1$ gravity that conical defect is the exterior geometry of a point mass.

[a] Lecture timestamp: 00:00:53–00:01:04.

Let δ denote the deficit angle. The lecture needs only the proportionality

$$\delta \propto G_3 m, \tag{1.2}$$

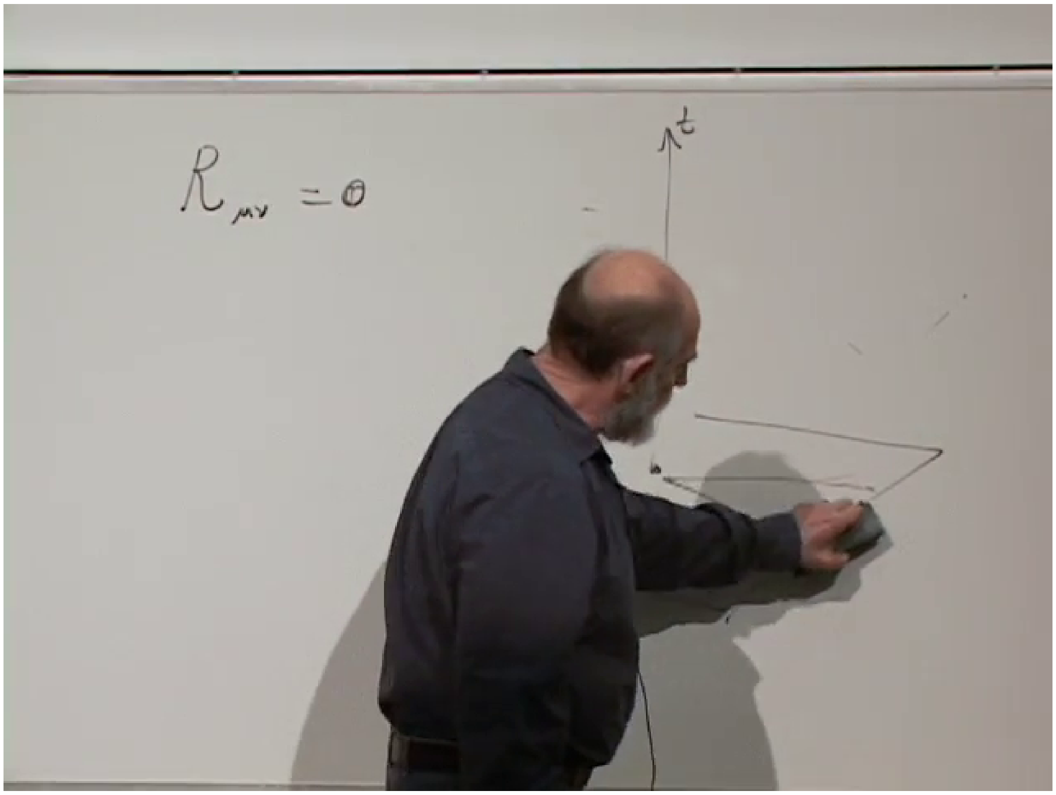

Figure 1.1: The conical exterior of a point source in two spatial dimensions. The board sketch shows the wedge construction while time is understood to extend perpendicular to each conical spatial slice.

Lecture frame: 00:01:45.

where G_3 is the Newton constant appropriate to three spacetime dimensions. The precise coefficient depends on convention, so the important statement here is simpler: no mass means no missing wedge, and increasing the mass increases the deficit.

The ideal point is not essential. Spread the matter through a small bounded region and the tip becomes rounded inside that region. Outside it, however, the geometry is still the same cone determined by the total mass. Thus a vacuum region can be locally flat and nevertheless retain global information about a source enclosed by a loop.

Question from the audience. Does the deficit angle grow with the mass, and what happens if it reaches a full turn? [a]

Response. Yes. As the missing angle approaches 2π, the opening closes. In the normalization used in the lecture, this occurs when the mass reaches the scale set by the three-dimensional Planck mass. The humorous conclusion is also the physical one: a universe with only two spatial dimensions leaves very little room for elaborate gravitating structure.

[a] Lecture timestamp: 00:01:52–00:04:29.

One spatial dimension is still more restrictive. Three spatial dimensions are therefore not merely an aesthetic preference; the number of dimensions controls both the global geometry around sources and the stability of ordinary systems.

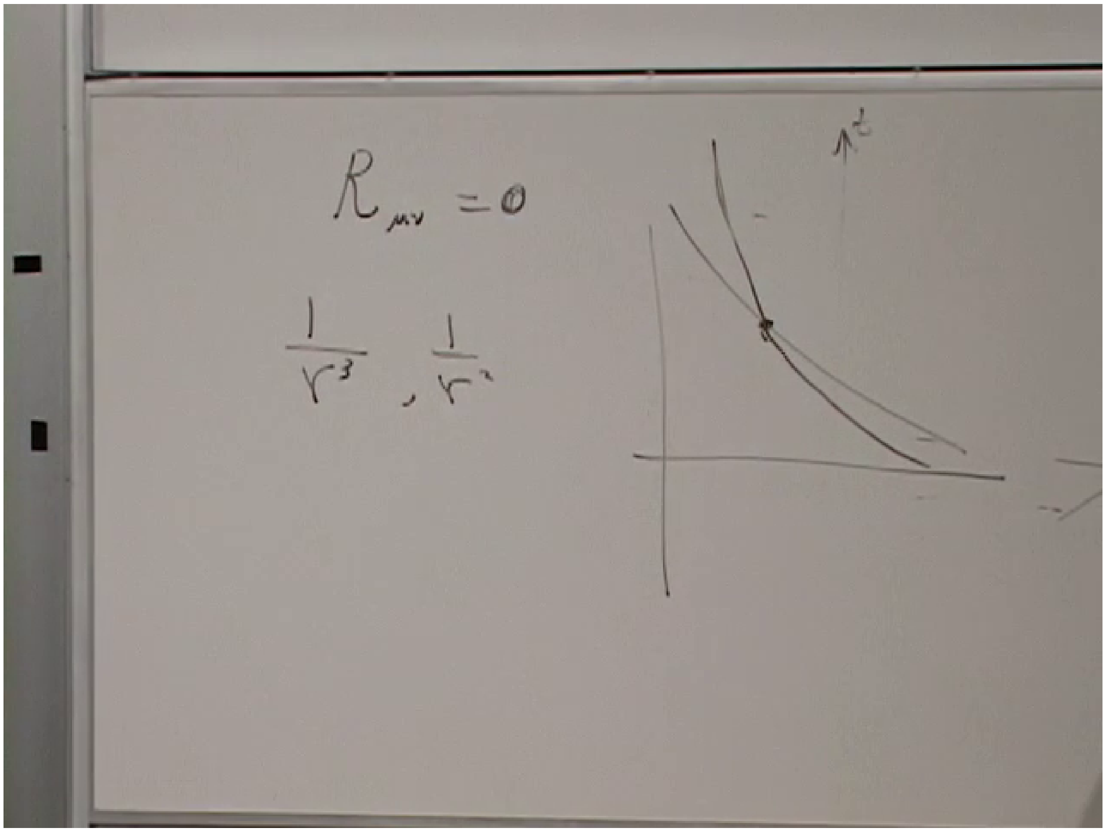

Figure 1.2: The board comparison of inverse-power force laws. Relative to $1/r^2$, a higher-dimensional force is weaker at large radius and more singular at short radius.

Lecture frame: 00:08:00.

1.2 Why Three Spatial Dimensions Work

Too few dimensions make gravity cramped. Too many make familiar force laws dangerously steep. In d spatial dimensions, Gauss's law gives the radial scaling

$$F_d(r) \propto \frac{1}{r^{d-1}}, \tag{1.3}$$

so the familiar inverse-square law in $d = 3$ becomes inverse-cube in $d = 4$, inverse-fourth-power in $d = 5$, and so on.

Question from the audience. If there are more spatial dimensions, does Newton's law remain inverse square? [a]

Response. No. The flux from a point source spreads over a $(d-1)$-sphere whose area grows as r^{d-1}. The force therefore falls as $1/r^{d-1}$.

[a] Lecture timestamp: 00:05:58–00:06:14.

The change is fatal to the systems on which ordinary complexity depends. At large radius the attraction loses its grip more quickly, so planetary orbits are not robust. At small radius the force becomes more singular, so an electron driven inward does not encounter the same stable balance that exists in three dimensions. Loosely speaking, the outer electronic structure escapes while inner states collapse toward the nucleus. Without stable atoms there is no ordinary chemistry.

Question from the audience. Could nuclear forces rescue matter even if electromagnetic atoms fail? [a]

Response. They face the same dimensional problem. Making a short-range interaction still more singular changes the binding of quarks and nuclei rather than preserving the hierarchy of scales we know. The argument is qualitative, but it shows how much of familiar physics is tied to three spatial dimensions.

[a] Lecture timestamp: 00:09:08–00:10:01.

Question from the audience. What about three spatial dimensions and two time dimensions? [a]

Response. That possibility is set aside rather than dressed up as a solved problem. Multiple time directions lead to difficult questions about evolution, causality, and stability. They are not needed for the argument being developed here.

[a] Lecture timestamp: 00:10:16–00:11:01.

No deep theorem has yet been offered that forces nature to choose $3 + 1$. The modest conclusion is that four-dimensional spacetime is empirically special and structurally hospitable.

Question from the audience. Why count time as a dimension at all? [a]

Response. Because special relativity does not preserve a universal split between space and time. Lorentz boosts mix t with a spatial coordinate, just as energy and momentum mix as components of a four-vector. Time is distinguished by the Lorentzian sign in Eq. (1.1), but it is not an external parameter standing outside spacetime. Different inertial observers choose different time axes through the same four-dimensional geometry.

[a] Lecture timestamp: 00:11:47–00:13:23.

1.3 The Conserved Geometric Tensor

An audience question now brings the cosmological constant into the room. Before asking what it does, we should see where it can enter. The source $T_{\mu\nu}$ is a symmetric rank-two tensor, so the geometric side must have the same character. From curvature, the simplest candidates are

$$R_{\mu\nu}, \qquad g_{\mu\nu}R. \tag{1.4}$$

Begin with the linear combination

$$R_{\mu\nu} + C\, g_{\mu\nu}R. \tag{1.5}$$

The coefficient is not chosen for visual symmetry. Local energy-momentum conservation requires

$$\nabla^\mu T_{\mu\nu} = 0, \tag{1.6}$$

and therefore the geometric tensor equated to it must be covariantly conserved identically. The contracted Bianchi identity fixes $C = -\frac{1}{2}$, giving

$$\boxed{G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R, \qquad \nabla^\mu G_{\mu\nu} = 0.} \tag{1.7}$$

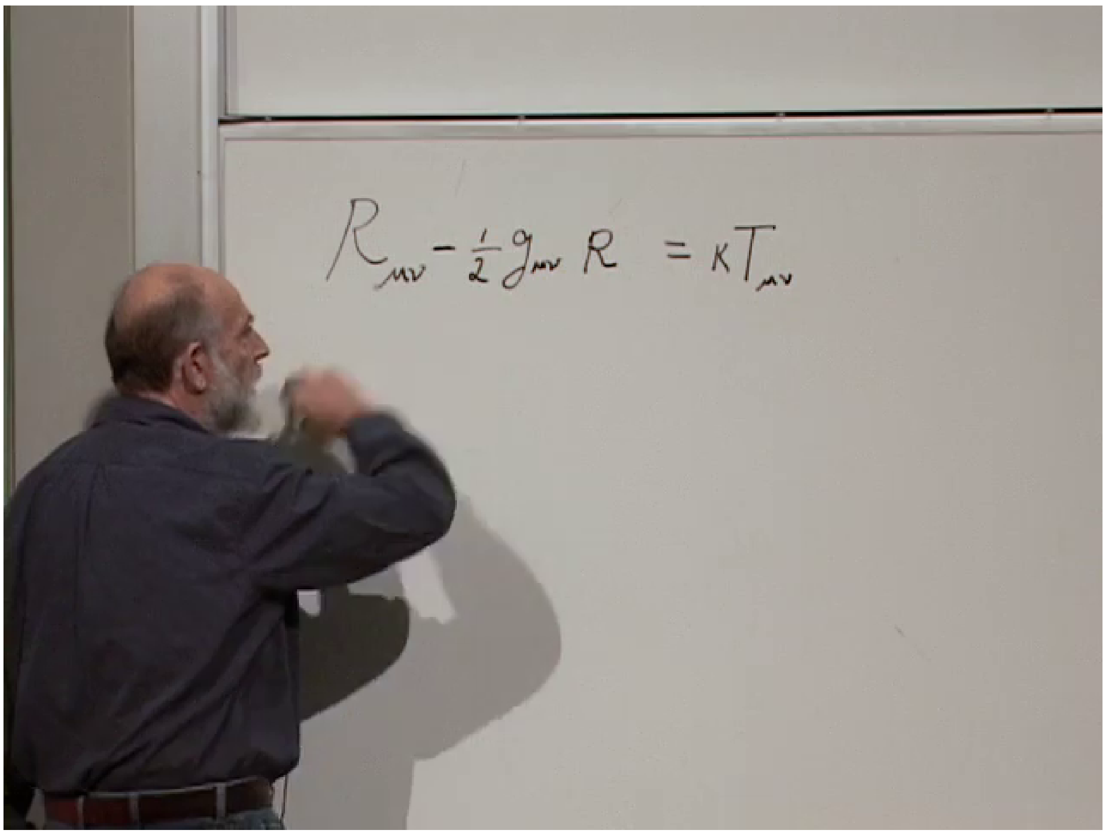

Figure 1.3: The Ricci tensor and scalar-curvature combination from which the conserved Einstein tensor is formed.

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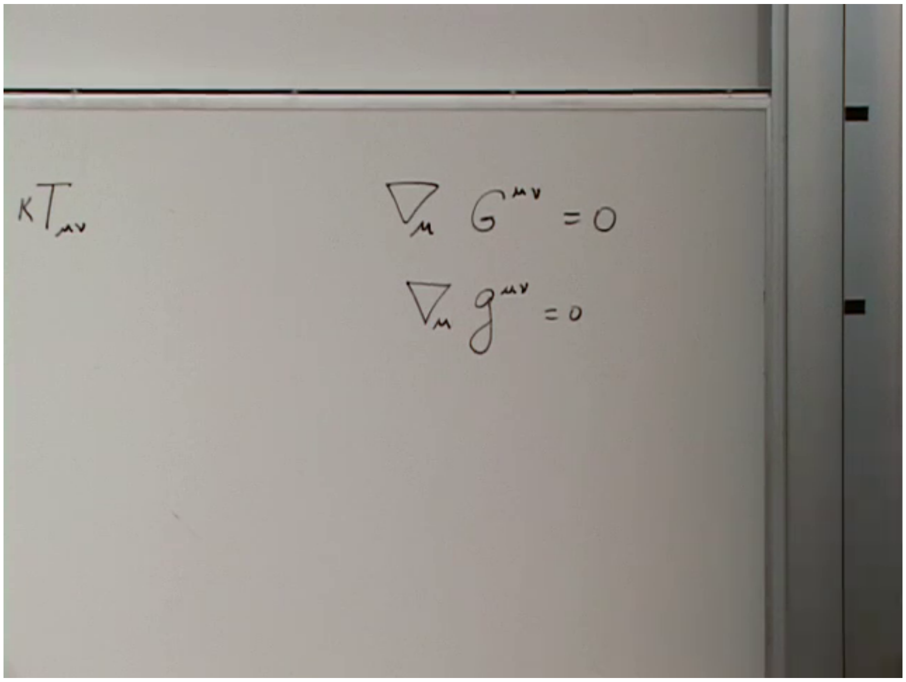

Figure 1.4: The two conservation identities paired on the board: $\nabla_\mu G^{\mu\nu} = 0$ and metric compatibility $\nabla_\mu g^{\mu\nu} = 0$.

Lecture frame: 00:17:15.

Metric compatibility supplies a second identically conserved rank-two tensor:

$$\nabla_\alpha g_{\mu\nu} = 0. \tag{1.8}$$

Consequently a constant multiple of the metric may be added without disturbing the conservation law. With the standard normalization, Einstein's equation is

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.} \tag{1.9}$$

Question from the audience. Why is the covariant derivative of the metric zero? [a]

Response. At any event choose local inertial coordinates. At that event

$$g_{\mu\nu} = \eta_{\mu\nu}, \qquad \partial_\alpha g_{\mu\nu} = 0,$$

and the Christoffel symbols vanish. Hence $\nabla_\alpha g_{\mu\nu} = 0$ there. Because the covariant derivative of the metric is itself a tensor, vanishing in one coordinate system at an arbitrary event means vanishing in every coordinate system at every event. The metric is not zero; its covariant derivative is.

[a] Lecture timestamp: 00:17:05–00:18:01.

In the Newtonian analogy, Λ does not merely correct an inverse-square law. It creates an acceleration proportional to distance. Its observed value is so small that this contribution is negligible on laboratory and planetary scales, yet it can dominate on cosmological scales. We postpone that consequence until the end of the lecture.

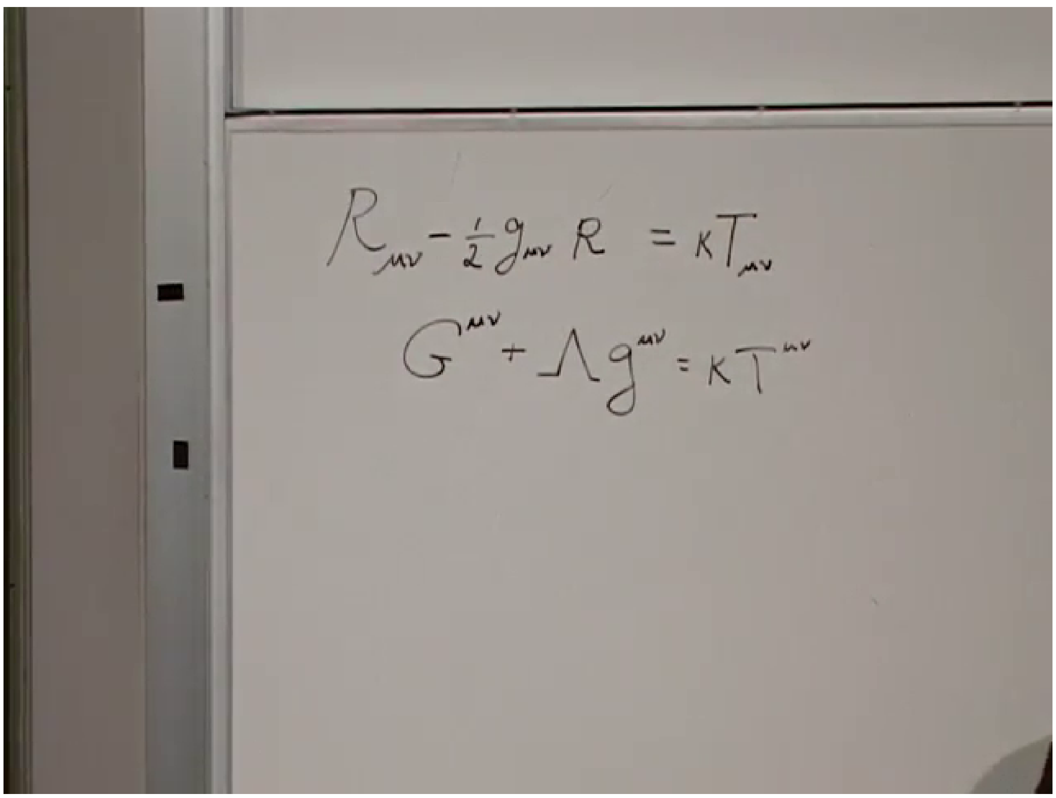

Figure 1.5: The cosmological term enters as a constant multiple of the metric, preserving the same covariant conservation law as the Einstein tensor.

Lecture frame: 00:19:00.

1.4 A Constrained Guess Tested Against Newton

Question from the audience. Was Einstein's equation derived, or was it guessed and then checked experimentally? [a]

Response. It was a highly constrained physical guess. Einstein demanded a tensor equation with the same form in every coordinate system, required local energy-momentum conservation, and required the weak, static, slow-motion limit to reproduce Newtonian gravity. Those conditions narrow the choice dramatically, but experiment still has the last word.

[a]Lecture timestamp: 00:20:41–00:25:16.

To see the correspondence, write a weak static metric as

$$g_{00} \simeq 1 + \frac{2\Phi}{c^2}, \qquad \left|\frac{\Phi}{c^2}\right| \ll 1. \tag{1.10}$$

For slowly moving matter, rest energy dominates:

$$T_{00} \simeq \rho c^2. \tag{1.11}$$

The 00-component of Einstein's equation then reduces to the Poisson equation

$$\nabla^2 \Phi = 4\pi G\rho \tag{1.12}$$

when Λ is neglected. Equivalently, the geometric side contains spatial second derivatives of g_{00}. The factor of two in Eq. (1.10) is precisely the bookkeeping detail the lecture warns us not to lose.

In weak fields and at low speeds this reproduces the older theory. At high speeds or in strong fields the two theories differ, and the relativistic equation is accepted because its new predictions survive observation. Covariance guides the guess; agreement with nature justifies it.

1.5 The Equivalence Principle as a Working Rule

Before solving for the exterior geometry of a star, return to the equivalence principle. Its modern use is local and operational. Near any event one can choose a freely falling frame in which

$$g_{\mu\nu}(P) = \eta_{\mu\nu}, \qquad \partial_\alpha g_{\mu\nu}(P) = 0. \tag{1.13}$$

Curvature, which depends on second derivatives and quadratic connection terms, cannot in general be removed. But over a sufficiently small laboratory the laws of physics take their special-relativistic form. The geographical analogy is a small patch of Earth's surface: it may be treated as flat even though the whole surface is curved.

This gives a recipe:

1. write the familiar flat-spacetime law in genuine tensor form;
2. replace the flat metric by the curved metric and ordinary derivatives by covariant derivatives where the tensor structure requires it;
3. interpret the resulting equation locally in freely falling frames and globally on the curved geometry.

For a free particle, straight inertial motion becomes geodesic motion. For the next example, use a scalar wave.

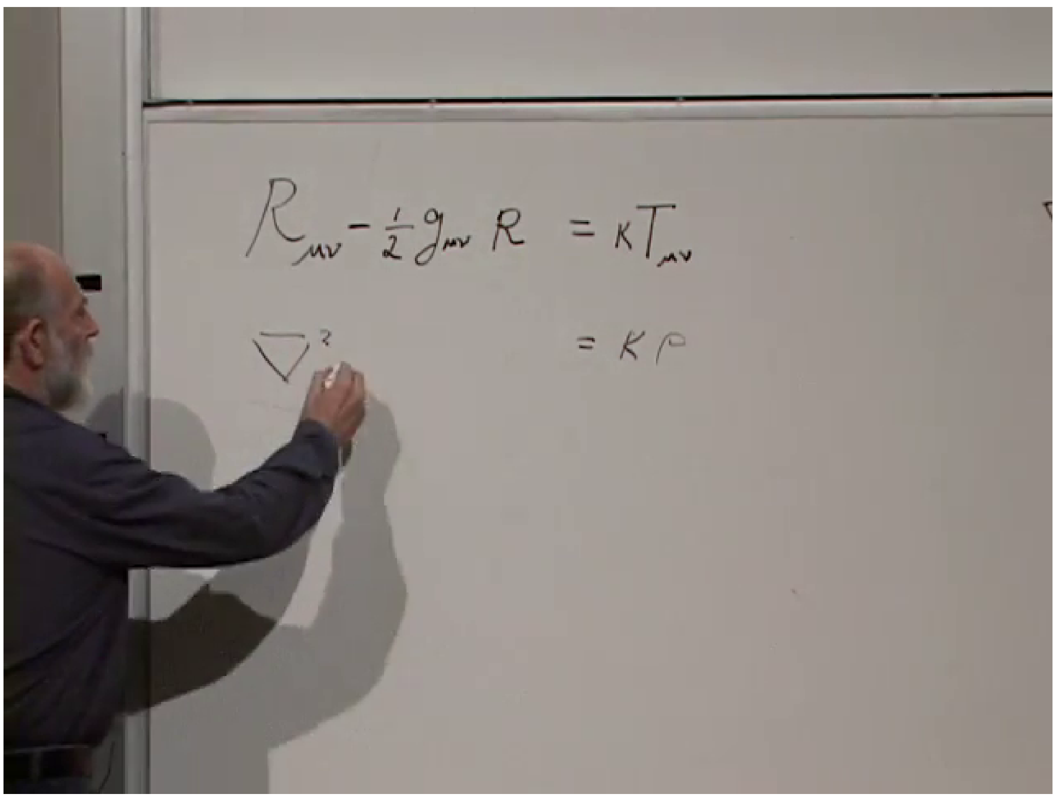

Figure 1.6: The 00-component is reduced on the board to a Poisson-type equation with energy density as its source.

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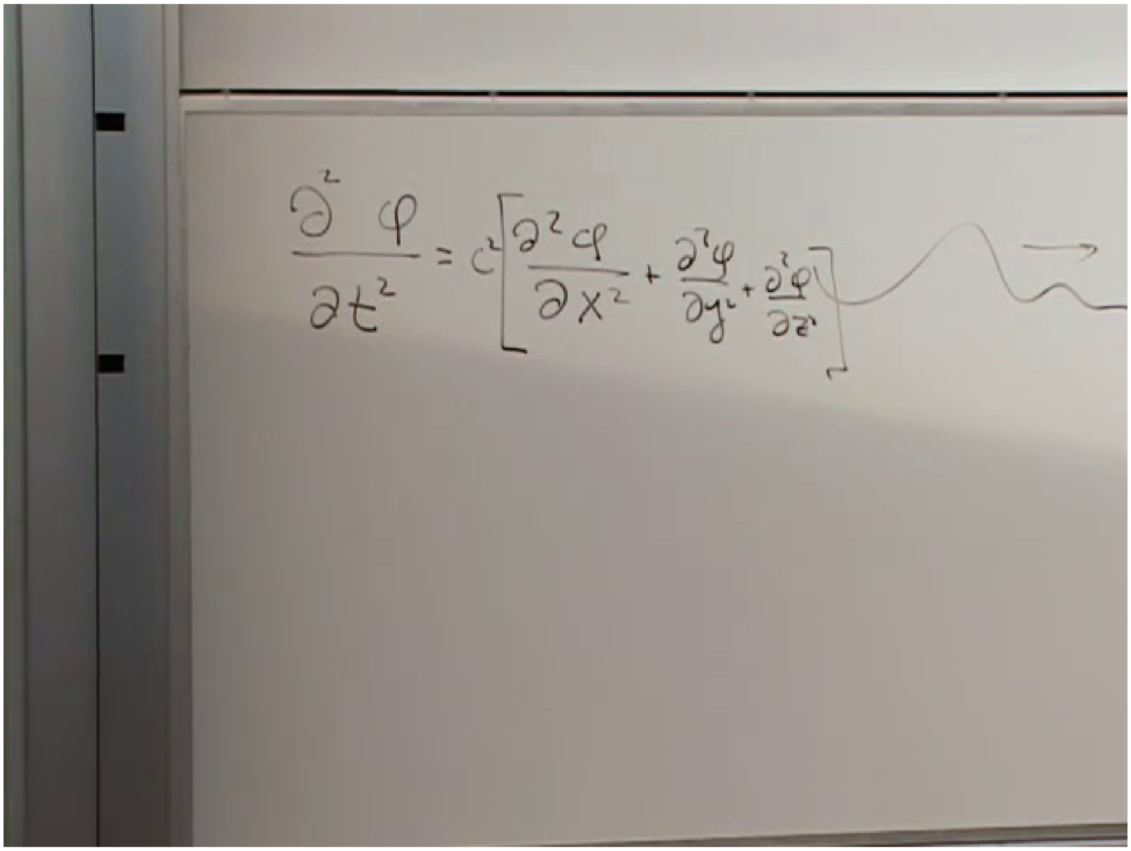

Figure 1.7: The one-dimensional wave equation extended by the two additional spatial derivatives, with a traveling profile sketched at the right.

Lecture frame: 00:34:15.

1.5.1 The flat-space wave

In one spatial dimension,

$$\frac{\partial^2\phi}{\partial t^2} = c^2\frac{\partial^2\phi}{\partial x^2}. \tag{1.14}$$

Its general motion is assembled from profiles traveling left and right,

$$\phi(x,t) = f(x-ct) + g(x+ct), \tag{1.15}$$

and linear combinations remain solutions. In three spatial dimensions,

$$\frac{\partial^2\phi}{\partial t^2} = c^2\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}\right). \tag{1.16}$$

Setting $c=1$, the same equation becomes

$$\eta^{\mu\nu}\partial_\mu\partial_\nu\phi = 0. \tag{1.17}$$

The compact notation creates a useful trap. Replacing $\eta^{\mu\nu}$ by $g^{\mu\nu}(x)$ gives an expression that looks covariant,

$$g^{\mu\nu}\partial_\mu\partial_\nu\phi, \tag{1.18}$$

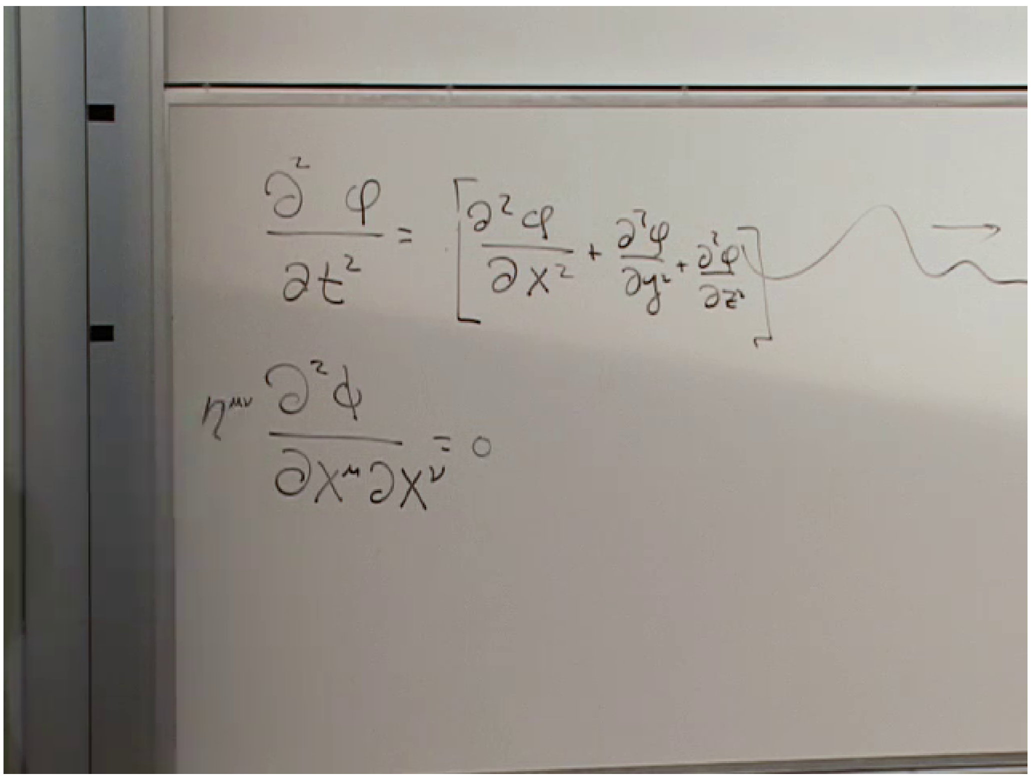

Figure 1.8: The flat-space scalar wave equation written in compact Minkowski form beneath its component expansion.

Lecture frame: 00:35:15.

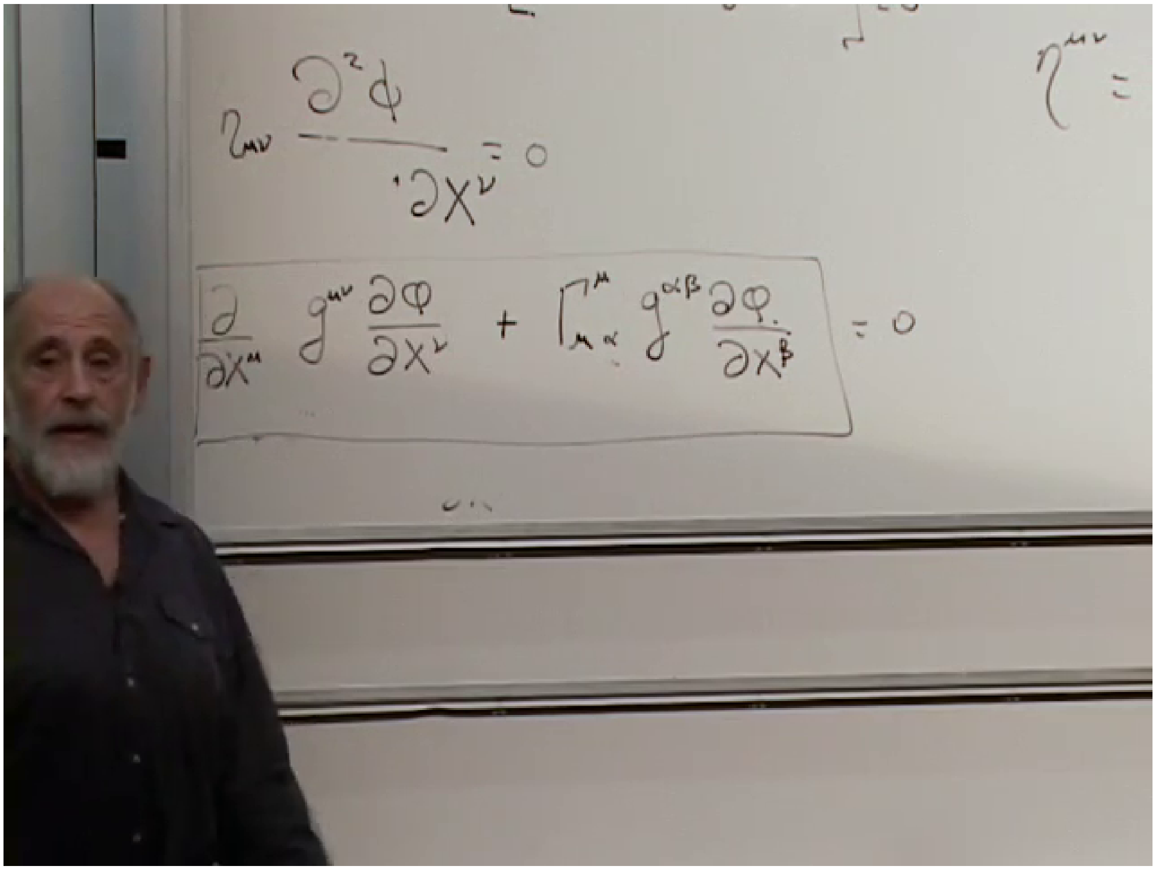

Figure 1.9: The curved-space wave equation written as the covariant divergence of the raised gradient. The connection term appears because a vector is being differentiated.

Lecture frame: 00:44:15.

but appearances are not enough.

Question from the audience. Why is $g^{\mu\nu}\partial_\mu\partial_\nu\phi = 0$ not already a tensor equation? [a]

Response. The first derivative $\partial_\nu\phi$ is a covector because ϕ is a scalar. Raising its index produces the vector $g^{\mu\nu}\partial_\nu\phi$. An ordinary derivative of that vector is not a tensor: derivatives of the coordinate transformation remain. The second differentiation must therefore be covariant.

[a]Lecture timestamp: 00:39:06–00:40:07.

The minimally coupled curved-spacetime equation is

$$\boxed{\nabla_\mu(g^{\mu\nu}\partial_\nu\phi) = \nabla_\mu\nabla^\mu\phi = 0.} \tag{1.19}$$

In explicit coordinates this is

$$\Box_g\phi = \frac{1}{\sqrt{-g}}\,\partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\big) = 0, \tag{1.20}$$

where $g = \det(g_{\mu\nu})$.

Question from the audience. Is general covariance by itself the whole equivalence principle? [a]

Response. No. Covariance says that the equations retain their tensorial meaning under a change of coordinates. The equivalence principle adds the physical statement that, in a sufficiently small freely falling laboratory, nongravitational physics reduces to special relativity. It is this local statement that motivates the minimal-coupling recipe.

[a] Lecture timestamp: 00:48:28–00:49:07.

Question from the audience. Does a covariant derivative obey the ordinary product rule? [a]

Response. Yes. It is a derivation:

$$\nabla_\mu(AB) = (\nabla_\mu A)B + A(\nabla_\mu B),$$

with the appropriate index action on each factor. Hence

$$\nabla_\mu(g^{\mu\nu}\partial_\nu\phi) = (\nabla_\mu g^{\mu\nu})\partial_\nu\phi + g^{\mu\nu}\nabla_\mu\partial_\nu\phi.$$

Metric compatibility kills the first term and leaves $g^{\mu\nu}\nabla_\mu\nabla_\nu\phi$.

[a] Lecture timestamp: 00:49:13–00:51:05.

The metric and connection now appear as position-dependent coefficients in the wave equation. The analogy is a wave moving through a medium whose density or dielectric response varies from place to place: its propagation changes and its rays bend. No detailed frequency claim is needed. The secure point is that geometry tells the wave how to propagate.

1.6 The Wave Must Gravitate in Return

The equation $\Box_g\phi = 0$ describes one direction of the interaction: geometry acts on the field. General relativity requires the reverse direction as well. The scalar wave carries energy and momentum, so it must supply a stress-energy tensor on the right-hand side of Einstein's equation.

Work first in flat spacetime, where

$$\Box\phi = \eta^{\mu\nu}\partial_\mu\partial_\nu\phi = 0. \tag{1.21}$$

The tensor must have two indices and must be built locally from the field. The first derivatives provide the indexed ingredients. A natural ansatz is

$$T_{\mu\nu} = A\,\partial_\mu\phi\,\partial_\nu\phi + C\,\eta_{\mu\nu}\partial_\alpha\phi\,\partial^\alpha\phi. \tag{1.22}$$

Question from the audience. Why use expressions quadratic in the field rather than terms linear in ϕ? [a]

Response. Changing ϕ to $-\phi$ should not reverse the sign of the energy. A crest and a trough, or the two sides of a harmonic oscillation, carry the same positive energy. The violin-string analogy makes the same point: both velocity energy and stretching energy are quadratic.

[a] Lecture timestamp: 00:56:50–00:59:47.

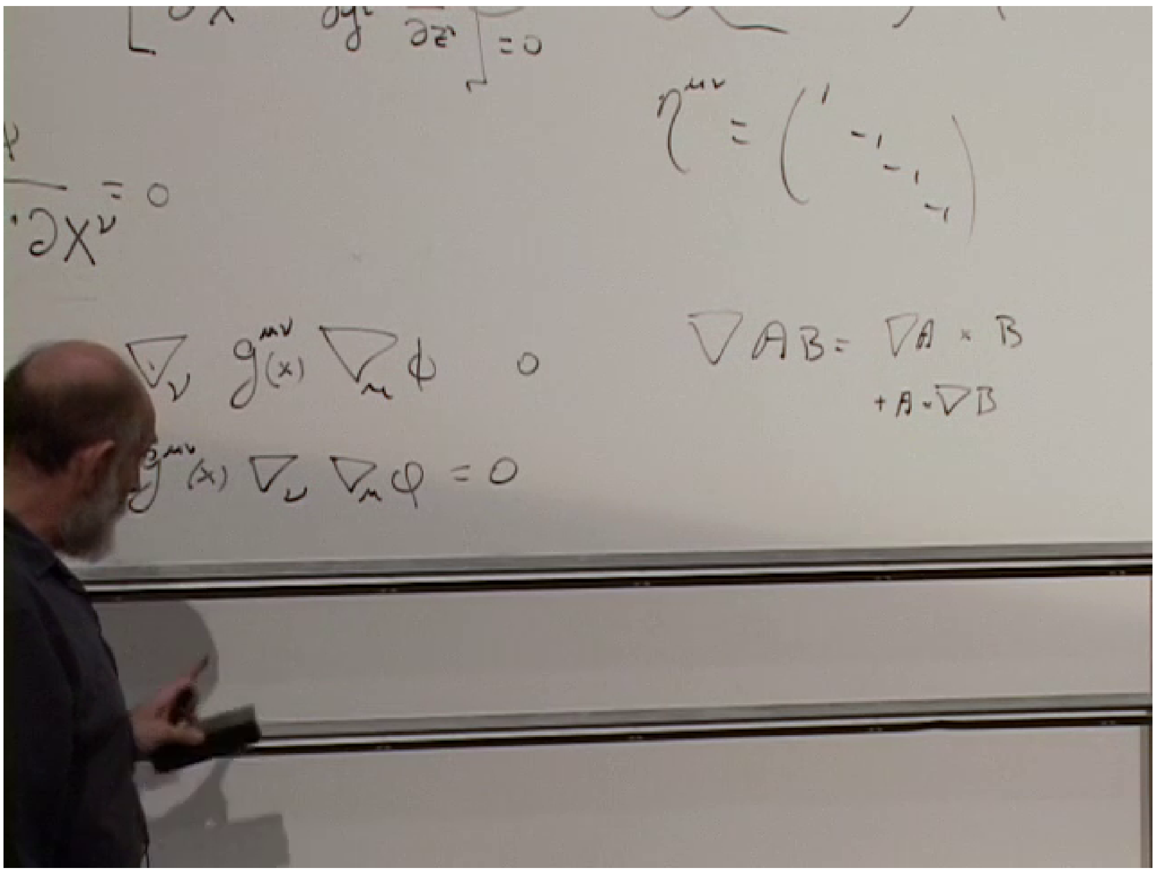

Figure 1.10: The product rule and metric compatibility reduce the covariant divergence to the curved d'Alembertian.

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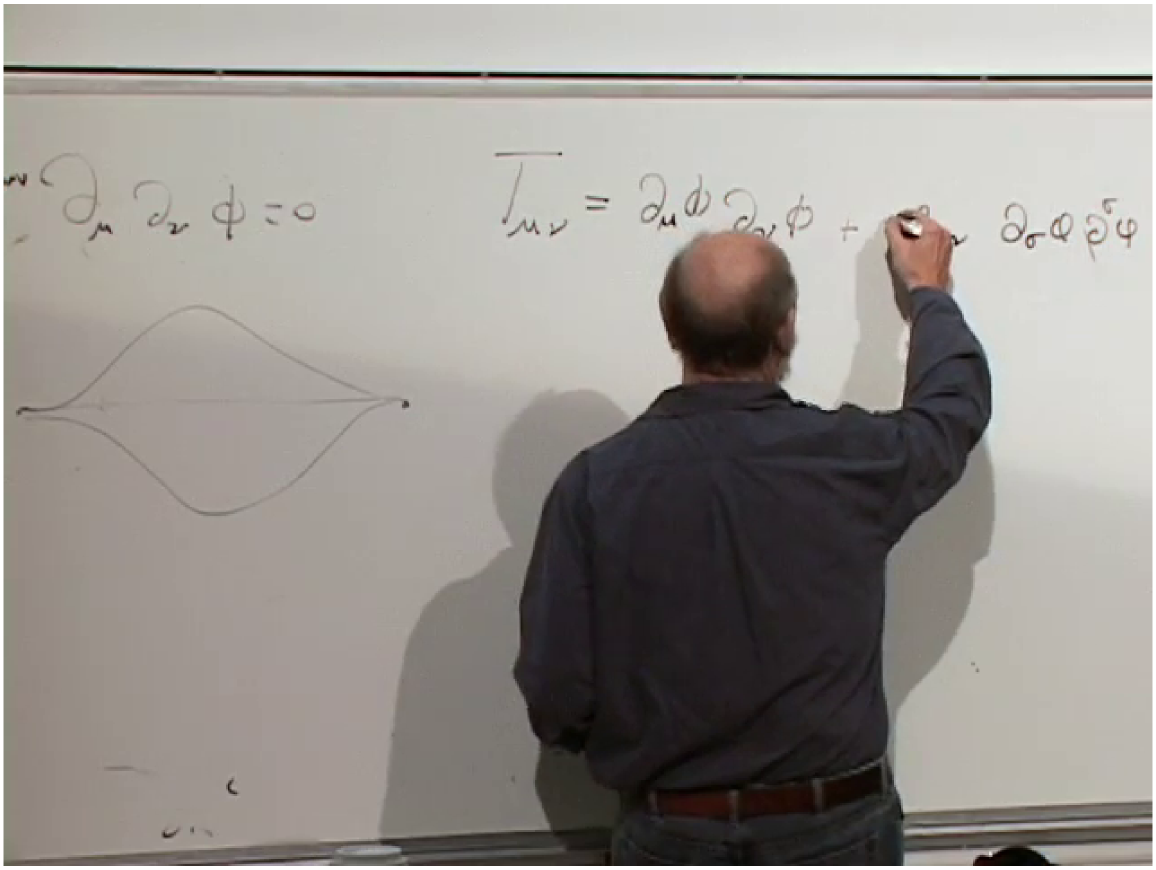

Figure 1.11: The quadratic scalar-field ansatz with the relative coefficient left undetermined. The neighboring sketches recall that both signs of a wave displacement carry the same energy.

Lecture frame: 01:00:15.

An overall normalization can be absorbed into a rescaling of ϕ, so set $A = 1$. Conservation will determine the relative coefficient C:

$$\partial^\mu T_{\mu\nu} = 0. \tag{1.23}$$

Question from the audience. What does the raised index on ∂^μ mean? [a]

Response. It is shorthand for contraction with the inverse Minkowski metric:

$$\partial^\mu = \eta^{\mu\alpha}\partial_\alpha.$$

Thus the conservation law is a four-divergence. Raising the index changes the signs of the spatial derivative components under the $(+---)$ convention.

[a] Lecture timestamp: 01:01:10–01:02:31.

Apply the product rule to the first term:

$$\partial^\mu \left(\partial_\mu\phi\, \partial_\nu\phi\right) = (\partial^\mu\partial_\mu\phi)\partial_\nu\phi + (\partial^\mu\phi)(\partial_\mu\partial_\nu\phi). \tag{1.24}$$

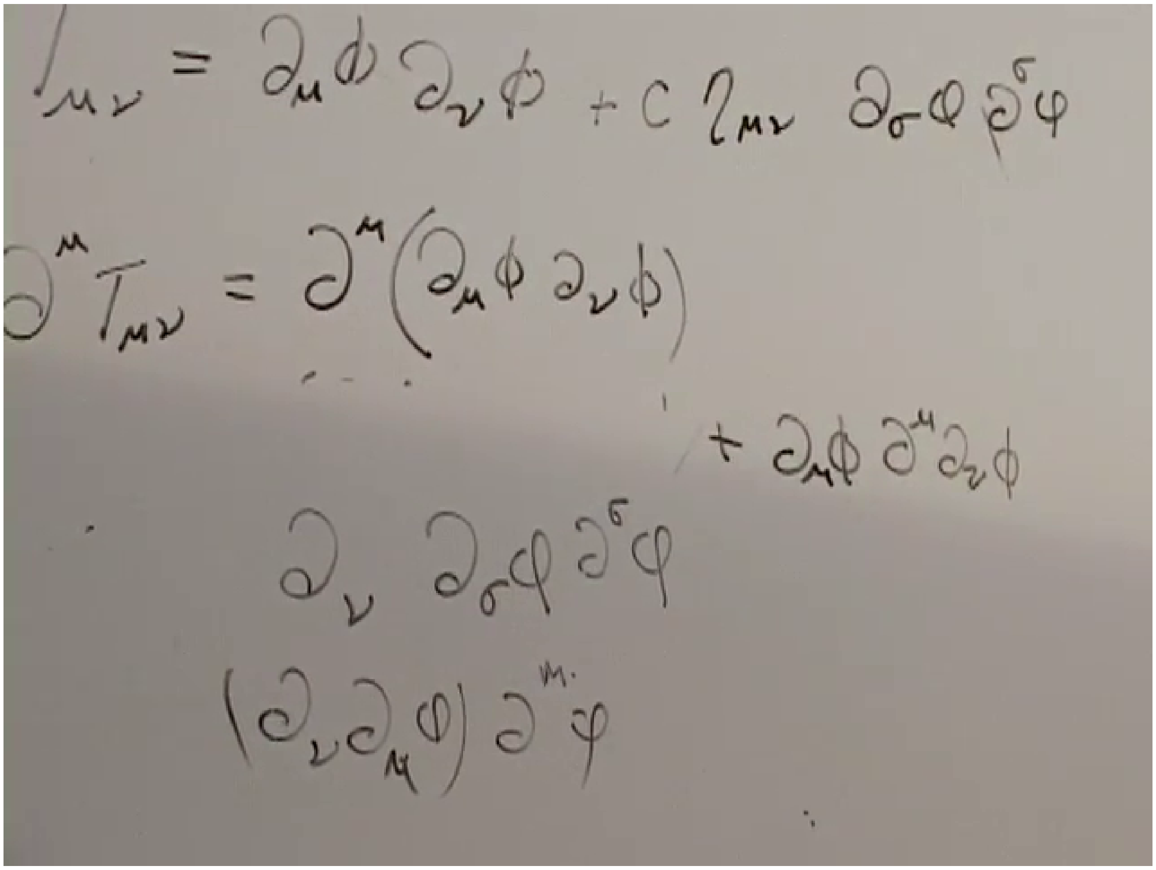

Figure 1.12: The product-rule calculation of $\partial^\mu T_{\mu\nu}$. One term vanishes by the wave equation; the two remaining derivative structures determine the relative coefficient.

Lecture frame: 01:06:30.

The first contribution vanishes by Eq. (1.21). For the second part of the ansatz,

$$\begin{aligned}\partial^\mu\left(\eta_{\mu\nu}\partial_\alpha\phi\,\partial^\alpha\phi\right) &= \partial_\nu\left(\partial_\alpha\phi\,\partial^\alpha\phi\right)\\ &= 2(\partial^\alpha\phi)(\partial_\nu\partial_\alpha\phi).\end{aligned} \tag{1.25}$$

Ordinary partial derivatives commute, and the dummy index can be renamed. The remaining term in Eq. (1.24) therefore has exactly the same form as one half of Eq. (1.25).

Question from the audience. Where did the first term go, and what fixes the minus sign? [a]

Response. The first term contains $\partial^\mu\partial_\mu\phi = \Box\phi$, so it vanishes on a solution of the wave equation. The surviving derivative of $\partial_\mu\phi\,\partial_\nu\phi$ must cancel the derivative of the metric-proportional term. Because the latter produces a factor of two, conservation requires $C = -\frac{1}{2}$.

[a] Lecture timestamp: 01:06:19–01:07:24.

We obtain

$$\boxed{T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}\partial_\alpha\phi\,\partial^\alpha\phi.} \tag{1.26}$$

This is the canonical stress-energy tensor for a free massless real scalar field in flat spacetime.

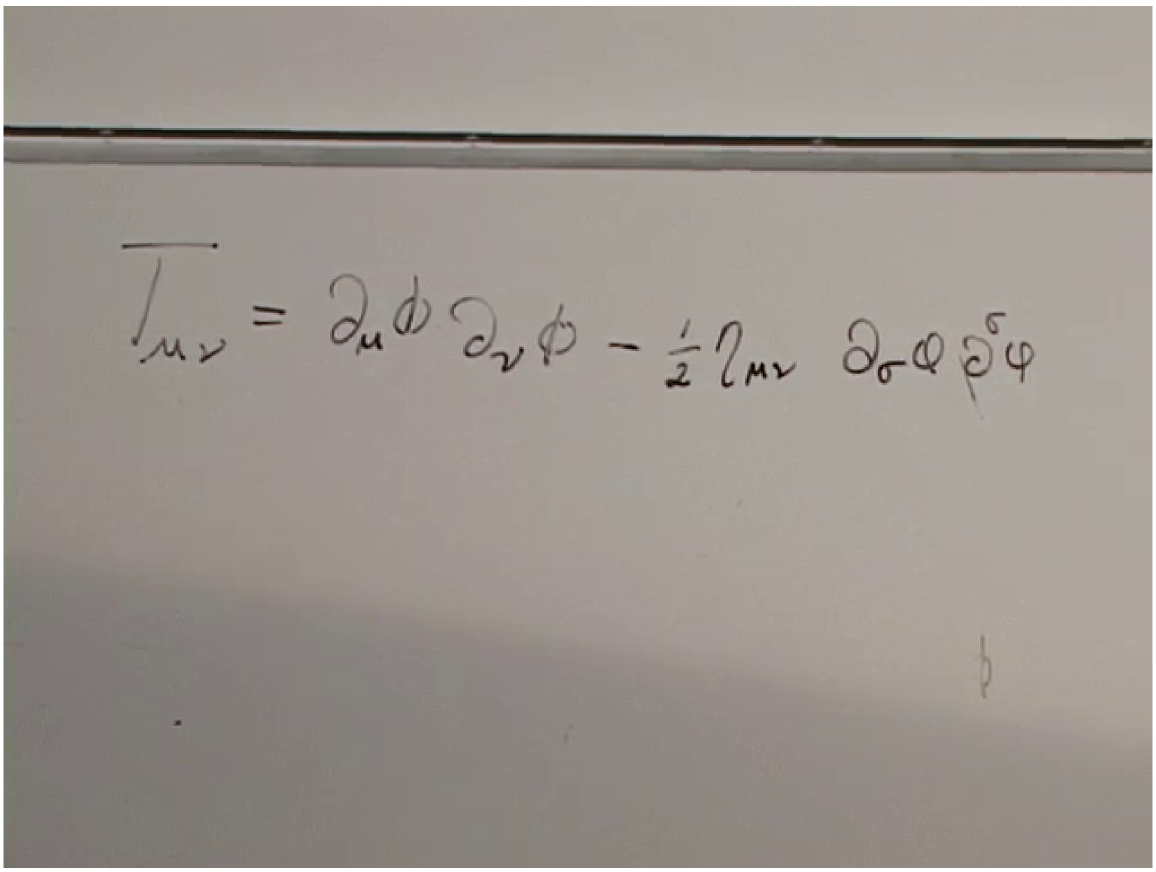

Figure 1.13: The completed scalar-field stress-energy tensor after the conservation condition fixes the coefficient $-\frac{1}{2}$.

Lecture frame: 01:08:15.

1.6.1 Reading the energy density

For $\mu = \nu = 0$, use

$$\partial_\alpha \phi \, \partial^\alpha \phi = \dot{\phi}^2 - |\nabla \phi|^2. \tag{1.27}$$

Then

$$\begin{aligned} T_{00} &= \dot{\phi}^2 - \frac{1}{2}\left(\dot{\phi}^2 - |\nabla\phi|^2\right) \\ &= \frac{1}{2}\left[\dot{\phi}^2 + \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2 + \left(\frac{\partial \phi}{\partial z}\right)^2\right]. \end{aligned} \tag{1.28}$$

Question from the audience. Does it matter whether the two zero indices are written upstairs or downstairs? [a]

Response. Not for this component in a local Minkowski frame with $\eta_{00} = 1$: raising both zero indices introduces no sign change. Spatial components require more care because each raised spatial index carries a minus sign.

[a] Lecture timestamp: 01:08:05–01:08:35.

The two positive terms have the familiar mechanics of a vibrating string. Motion of the string supplies kinetic energy; spatial bending or stretching supplies potential energy. A wave carries both. The calculation has therefore done more than satisfy an index identity: it has recovered the energy density we expected on physical grounds.

In curved spacetime the minimally coupled expression becomes

$$T_{\mu\nu} = \nabla_\mu \phi \, \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla_\alpha \phi \, \nabla^\alpha \phi, \tag{1.29}$$

and it is inserted together with $\Box_g \phi = 0$ into Eq. (1.9). Geometry changes the wave, and the wave's energy changes the geometry.

1.7 Pressure and Shear Also Gravitate

The phrase energy density can tempt us to look only at T_{00}, but Einstein's equation couples to the whole tensor. In a local orthonormal rest frame, an isotropic medium has

$$T_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}. \tag{1.30}$$

With one index raised, the same convention reads

$$T^{\hat{\mu}}{}_{\hat{\nu}} = \mathrm{diag}(\rho, -p, -p, -p). \tag{1.31}$$

Writing both forms prevents a common sign confusion.

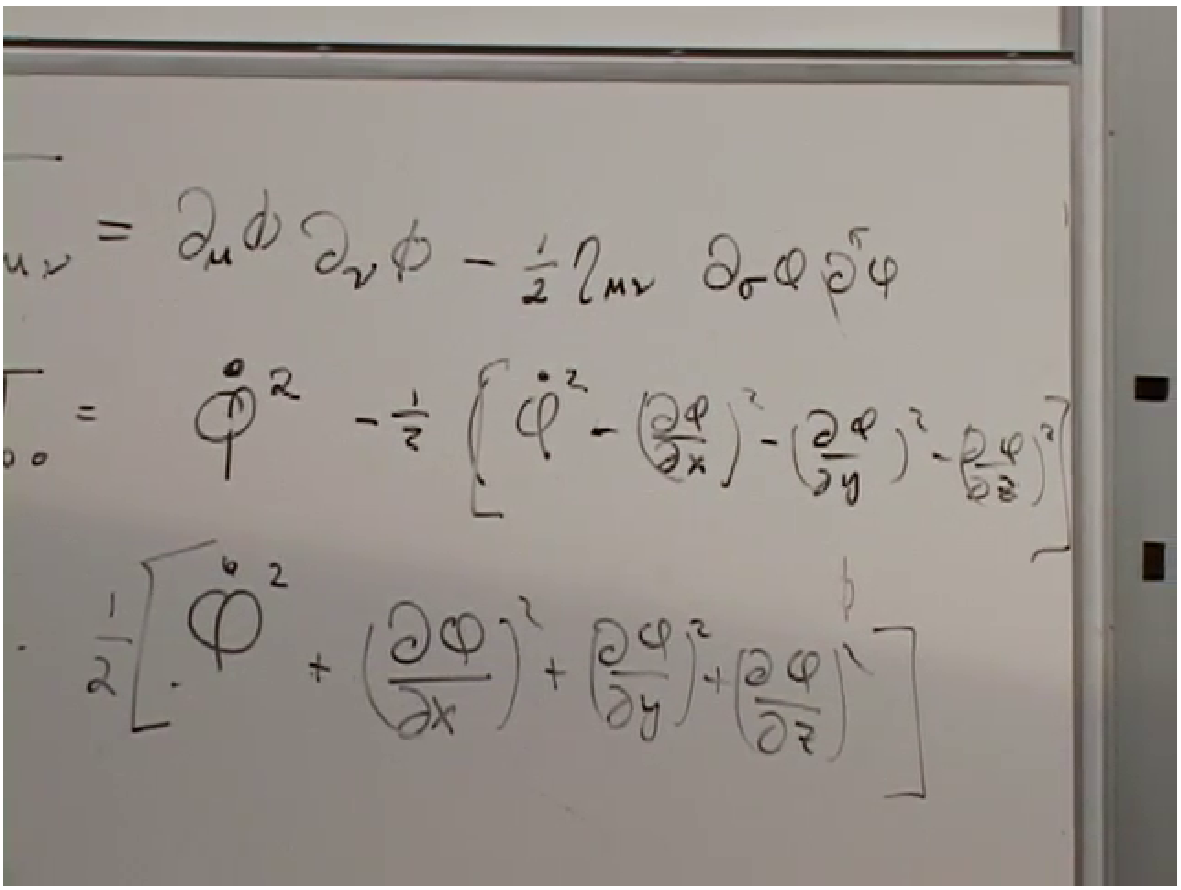

Figure 1.14: The stress tensor and its 00-component expanded into kinetic energy density $\dot{\phi}^2/2$ and gradient energy density $|\nabla\phi|^2/2$.

Lecture frame: 01:11:45.

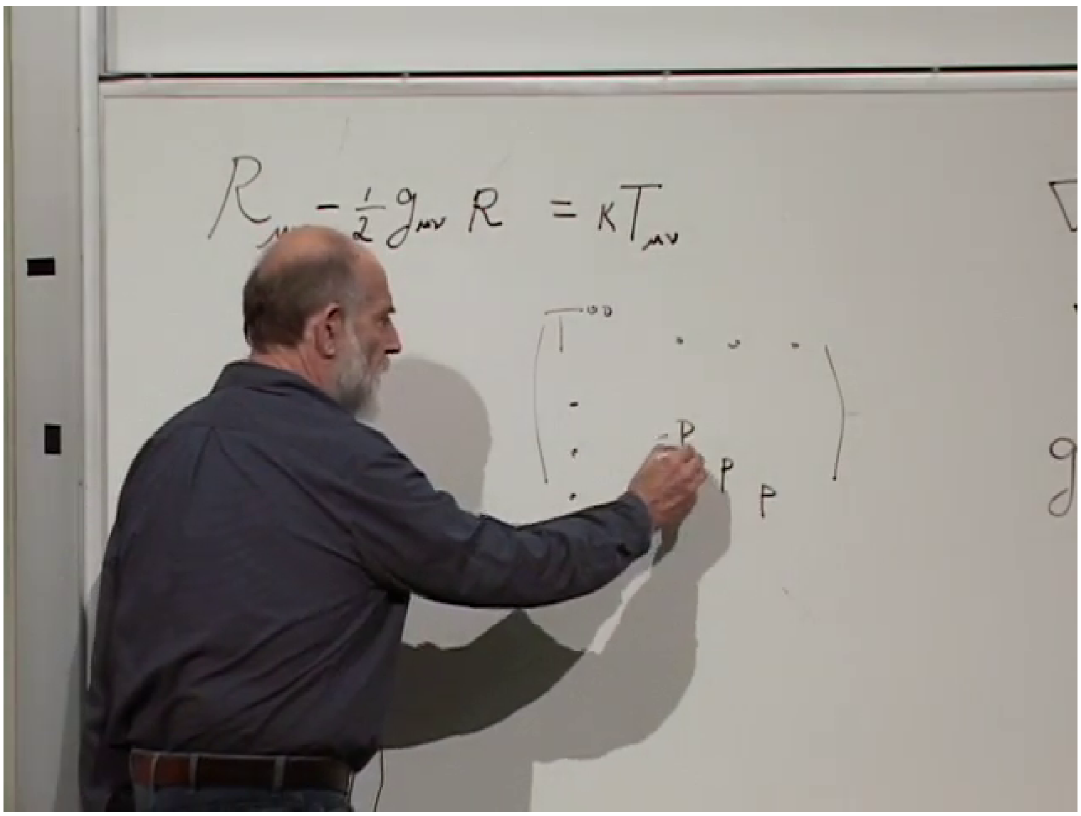

Figure 1.15: The isotropic rest-frame stress tensor on the board: energy density in the timelike slot and equal principal pressures in the three spatial directions.

Lecture frame: 01:15:45.

Question from the audience. Where does pressure appear in the stress-energy tensor? [a]

Response. In the spatial diagonal components in the fluid rest frame. For ordinary nonrelativistic matter, pressure is much smaller than the rest-energy density ρc^2, which is why Newtonian gravity can often ignore it. For relativistic radiation, pressure is of the same order as energy density; an isotropic radiation bath has $p = \rho/3$ when $c = 1$.

[a]Lecture timestamp: 01:14:09–01:16:01.

Off-diagonal spatial entries describe shear stress. Mixed time-space entries describe momentum density or energy flux. The tensor is symmetric in the systems under discussion, but symmetry does not force it to be diagonal in an arbitrary frame.

Question from the audience. Can pressure simply be added to the mass density as one more scalar source? [a]

Response. No. Pressure can be direction-dependent, and off-diagonal entries carry shear and momentum flux. Rotating the axes may diagonalize the spatial stress block, but its distinct eigenvalues remain. Gravity responds to the tensorial arrangement, not only to one number called equivalent mass.

[a]Lecture timestamp: 01:18:03–01:18:39.

The reciprocal structure can now be stated compactly:

$$\text{geometry} \xrightarrow{\Box_g \phi = 0} \text{field propagation}, \qquad \text{stress-energy} \xrightarrow{G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}} \text{geometry}. \tag{1.32}$$

In Wheeler's celebrated summary, matter tells spacetime how to curve, and curved spacetime tells matter how to move.

1.8 A Local Source Can Have a Nonlocal Field

Question from the audience. If a mass is localized, is $T_{\mu\nu}$ zero everywhere else, and should the gravitational field then vanish there? [a]

Response. The stress-energy tensor may indeed vanish outside the material body, but the field need not. The Newtonian analogue is Poisson's equation. Outside a compact source, $\rho = 0$, so $\nabla^2\Phi = 0$; nevertheless the solution $\Phi = -GM/r$ and its $1/r^2$ force remain nonzero. The source fixes boundary data that the vacuum solution carries outward.

[a]Lecture timestamp: 01:19:44–01:22:38.

The relativistic version requires one refinement. With $\Lambda = 0$, the exterior of an isolated body can satisfy

$$T_{\mu\nu} = 0, \qquad R_{\mu\nu} = 0, \tag{1.33}$$

without being flat. The full Riemann tensor may still be nonzero; its trace-free Weyl part carries the tidal field. Thus vacuum means Ricci-flat here, not necessarily Riemann-flat. This distinction is what allows the Schwarzschild exterior to be curved even though the matter lies inside the star.

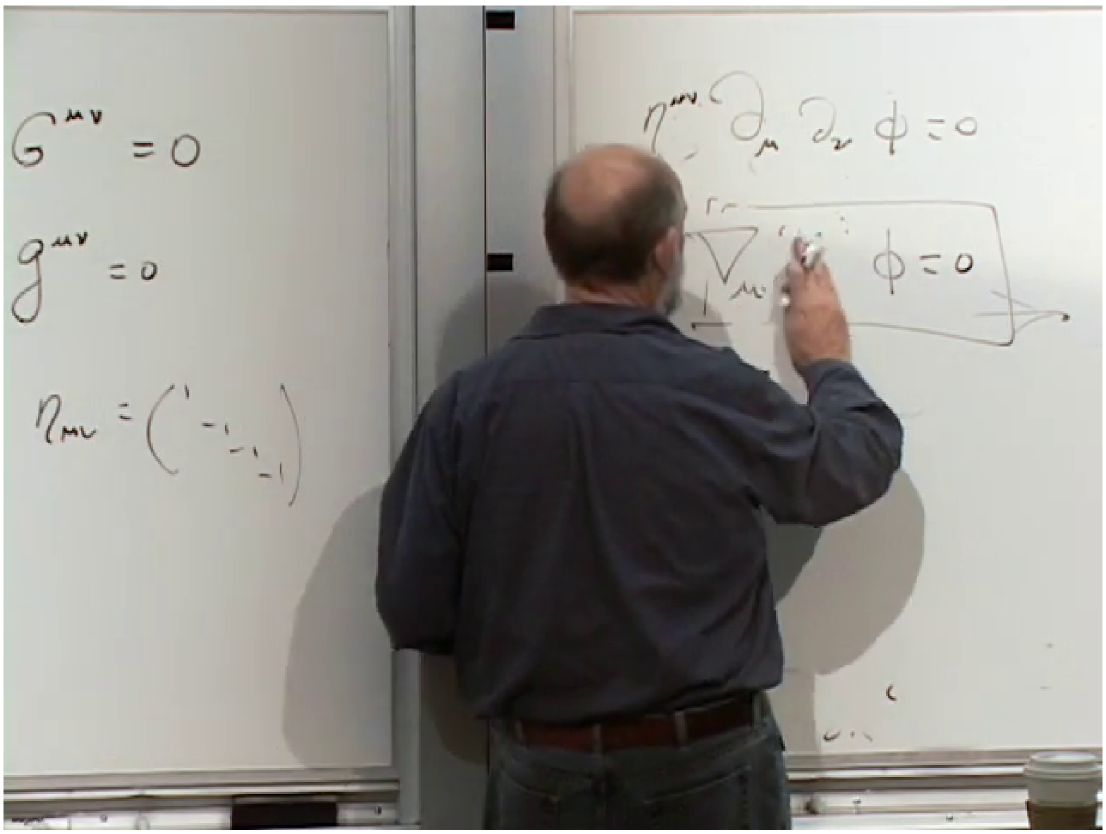

Figure 1.16: The two directions of the coupled system are kept together on the board: the wave equation contains geometry, while Einstein's equation contains the wave's stress-energy.

Lecture frame: 01:19:15.

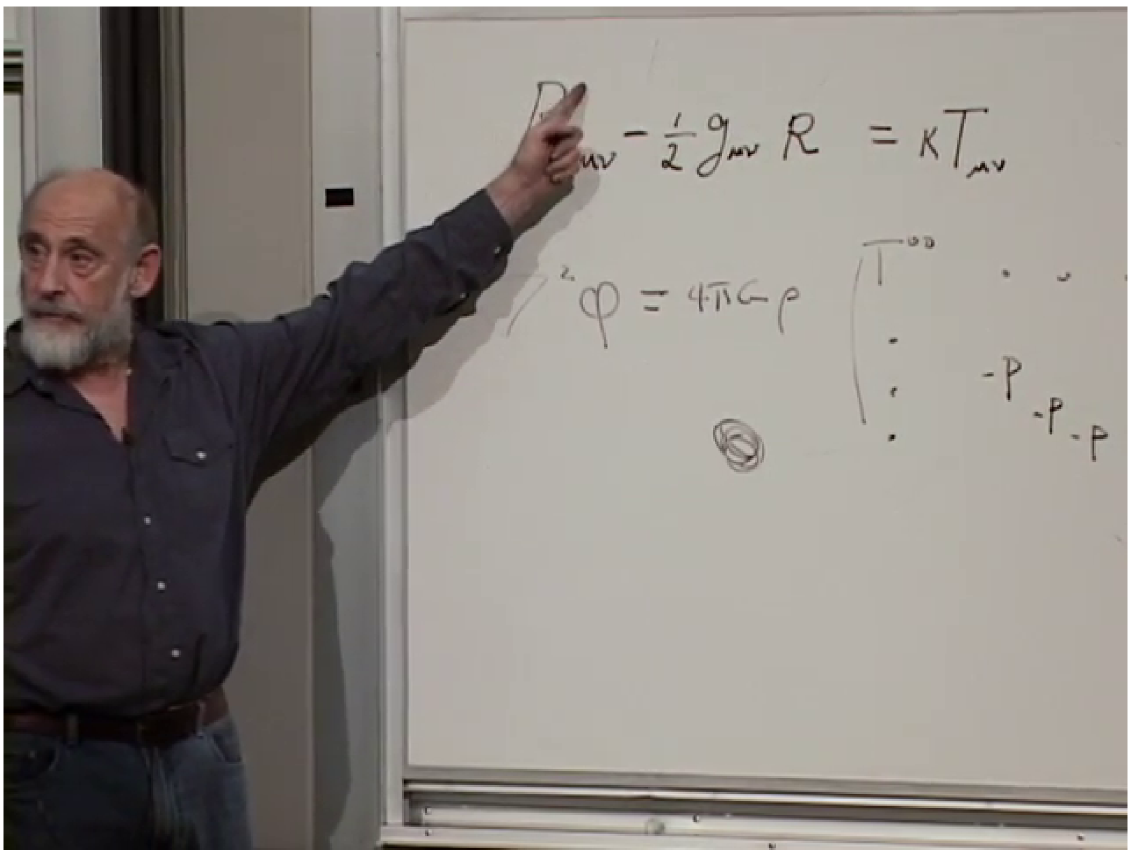

Figure 1.17: A localized source beside the Poisson equation. The density vanishes outside the source, while the exterior potential solves the homogeneous equation and remains nontrivial.

Lecture frame: 01:21:30.

Question from the audience. If light itself gravitates, does the superposition principle for waves fail? [a]

Response. In the coupled Einstein-field system, yes in principle. Two crossing or skew beams contribute stress-energy and therefore perturb the geometry through which each beam travels. The effect is ordinarily fantastically small because the gravitational mass associated with energy is E/c^2. A dimensional Newtonian estimate for two localized energy packets gives a scale like

$$F_{\text{grav}} \sim \frac{G}{r^2}\frac{E_1}{c^2}\frac{E_2}{c^2},$$

but this is only a magnitude estimate, not a substitute for the stress-energy of extended null beams.

[a] Lecture timestamp: 01:22:39–01:25:55.

Linear superposition is therefore an excellent approximation for ordinary light, not an exact principle of the full nonlinear theory.

1.9 The Cosmological Constant as Long-Range Gravity

Return now to Eq. (1.9). In the weak-field limit, the cosmological constant adds a uniform term to Poisson's equation. With our sign convention it may be written

$$\nabla^2\Phi = 4\pi G\rho - \Lambda c^2. \tag{1.34}$$

In empty space, a convenient spherically symmetric solution is

$$\Phi_\Lambda(r) = -\frac{\Lambda c^2}{6}r^2, \qquad \mathbf{a}_\Lambda = -\nabla\Phi_\Lambda = \frac{\Lambda c^2}{3}\mathbf{r}. \tag{1.35}$$

Positive Λ is repulsive in this convention; negative Λ is attractive.

Question from the audience. If the acceleration is proportional to distance from an origin, does the cosmological constant pick a preferred center? [a]

Response. No. The coordinate expression uses an origin, but the measurable statement concerns relative acceleration. For any two nearby points,

$$\mathbf{a}_\Lambda(\mathbf{r}_2) - \mathbf{a}_\Lambda(\mathbf{r}_1) = \frac{\Lambda c^2}{3}(\mathbf{r}_2 - \mathbf{r}_1),$$

which has the same form no matter where the coordinate origin is placed. A homogeneous expansion or contraction need not possess a central point.

[a] Lecture timestamp: 01:26:00–01:35:37.

Algebraically, Λ may be left with the geometry,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{1.36}$$

or moved to the matter side,

$$G_{\mu\nu} = 8\pi G\left(T_{\mu\nu} + T^{(\Lambda)}_{\mu\nu}\right), \qquad T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda}{8\pi G}g_{\mu\nu}, \tag{1.37}$$

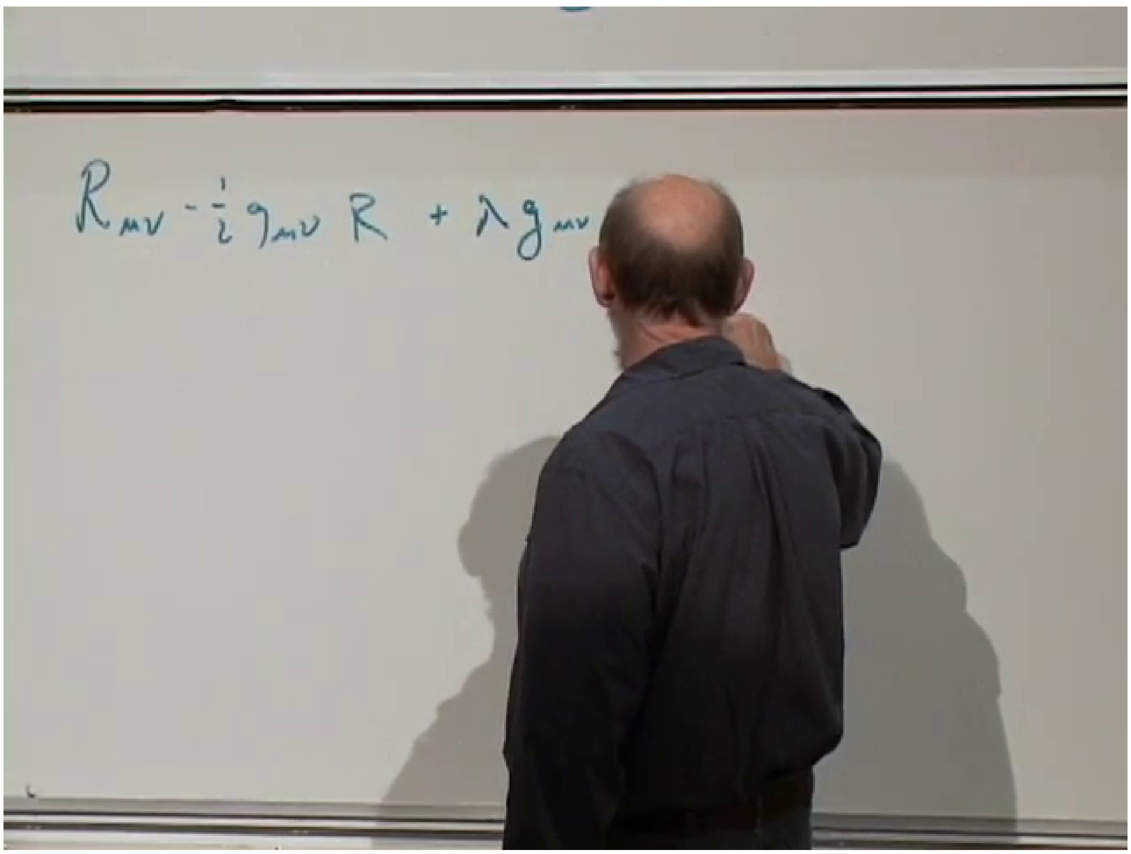

Figure 1.18: The Einstein equation with Λ and its Newtonian analogue: a constant source in Poisson's equation produces a quadratic potential.

Lecture frame: 01:27:15.

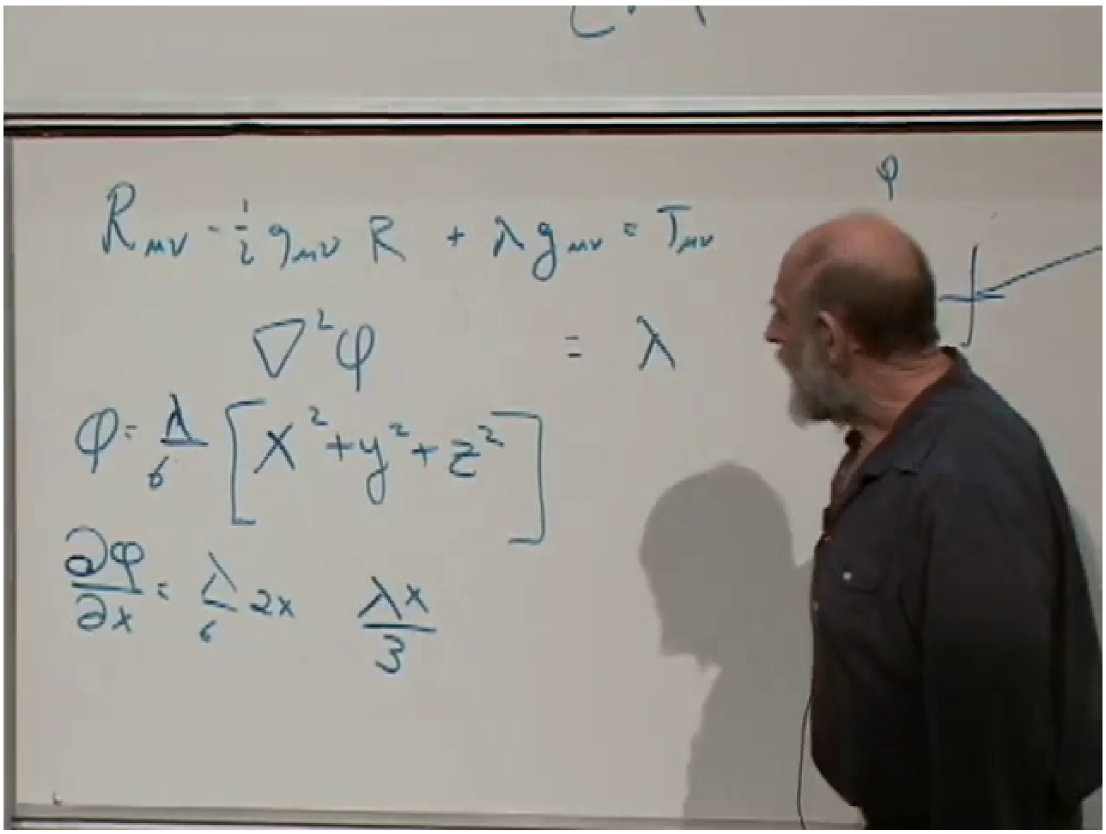

Figure 1.19: Differentiating the quadratic potential produces an acceleration linear in position. Relative accelerations are homogeneous, so the construction does not select a physical center.

Lecture frame: 01:32:15.

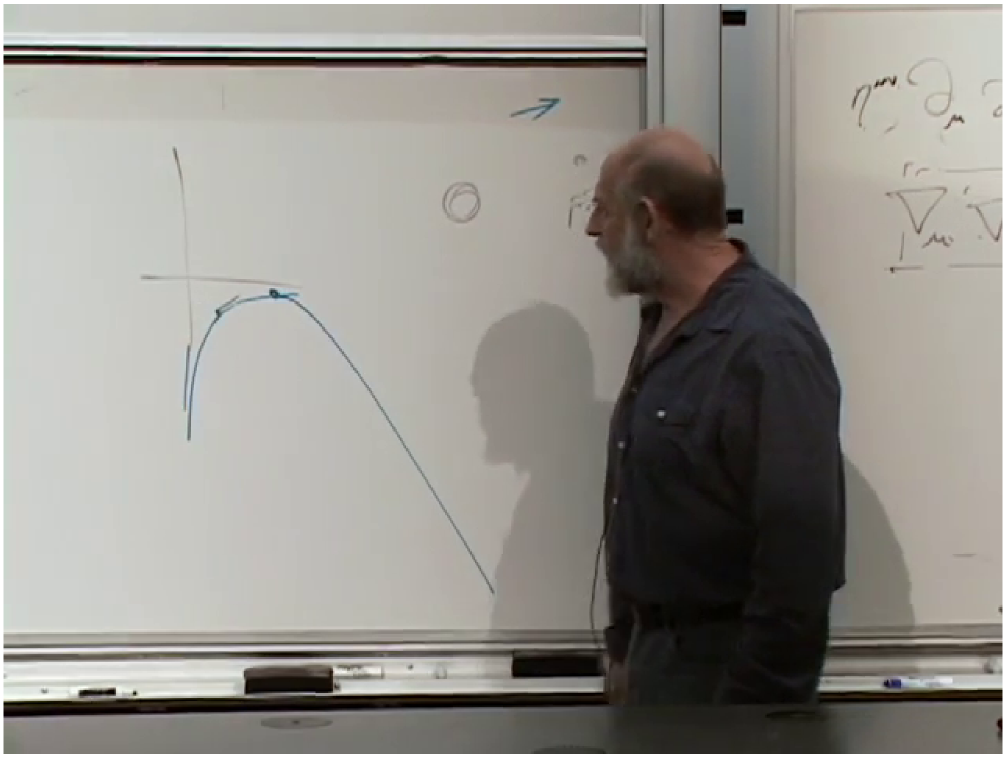

Figure 1.20: The effective-potential sketch for Einstein's static universe. The tuned static point sits at the top of the curve and is therefore unstable to either contraction or expansion.

Lecture frame: 01:39:30.

where $c = 1$. The second notation interprets the cosmological constant as vacuum stress-energy. It has energy density and pressure related by

$$p_\Lambda = -\rho_\Lambda. \tag{1.38}$$

Whether it is written as geometry or vacuum energy, the measurable equation is the same.

1.9.1 Einstein's unstable static universe

Historically, Einstein introduced a repulsive cosmological term to counter the mutual attraction of matter and hold the universe static. The cancellation can be tuned at one instant, but it is not a stable equilibrium.

Question from the audience. Why can the cosmological constant not hold a static universe in stable balance? [a]

Response. The lecture compares the balance to a rock placed at the top of a potential hill. Attraction and repulsion may cancel at one radius, but a small displacement does not produce a restoring force. Move slightly one way and attraction wins; move the other and repulsion wins. The stationary point is a maximum of the effective potential, not a minimum.

[a] Lecture timestamp: 01:35:37–01:40:29.

The mistake was not allowing the term in the field equation. The mistake was expecting a delicately balanced value to make a realistic static universe. A small positive Λ remains a legitimate and observationally important possibility.

Question from the audience. What would an attractive cosmological constant do? Would it create a preferred origin? [a]

Response. A negative Λ reverses the sign of the relative acceleration and favors contraction. It still does not select a center: every comoving observer can describe the same homogeneous relative motion around their own origin.

[a] Lecture timestamp: 01:40:32–01:41:23.

Question from the audience. Could the whole universe rotate? [a]

Response. No global rotation axis is observed. A rigidly rotating infinite universe would also force tangential speeds to grow without bound and eventually exceed c. Local regions may rotate, as galaxies do, but a universal rigid rotation would introduce a preferred axis and center absent from the homogeneous picture being discussed.

[a] Lecture timestamp: 01:41:24–01:43:00.

1.10 Principal Stresses and the Road to Schwarzschild

The audience returns once more to the off-diagonal entries of $T_{\mu\nu}$. In two spatial directions, a pure shear block may look like

$$S = \begin{pmatrix} 0 & \tau \\ \tau & 0 \end{pmatrix}. \tag{1.39}$$

Rotate the axes by 45°, and the same tensor becomes

$$S' = \begin{pmatrix} \tau & 0 \\ 0 & -\tau \end{pmatrix}. \tag{1.40}$$

What appeared as shear is now compression along one principal axis and tension along the other.

Question from the audience. What gravitational effect should one associate with off-diagonal stress? [a]

Response. It contributes to the full spacetime geometry just as the diagonal entries do. Time-dependent nonspherical stresses are among the ingredients that can generate gravitational radiation. For the static algebraic question, rotate to principal axes: shear becomes unequal pressures, making its physical content easier to see.

[a] Lecture timestamp: 01:43:10–01:45:37.

The next concrete geometry will be the exterior of a spherical mass. Symmetry reduces the task to finding a small number of radial metric functions, calculating their Christoffel symbols and curvature, and solving the vacuum Einstein equations. That solution is the Schwarzschild metric.

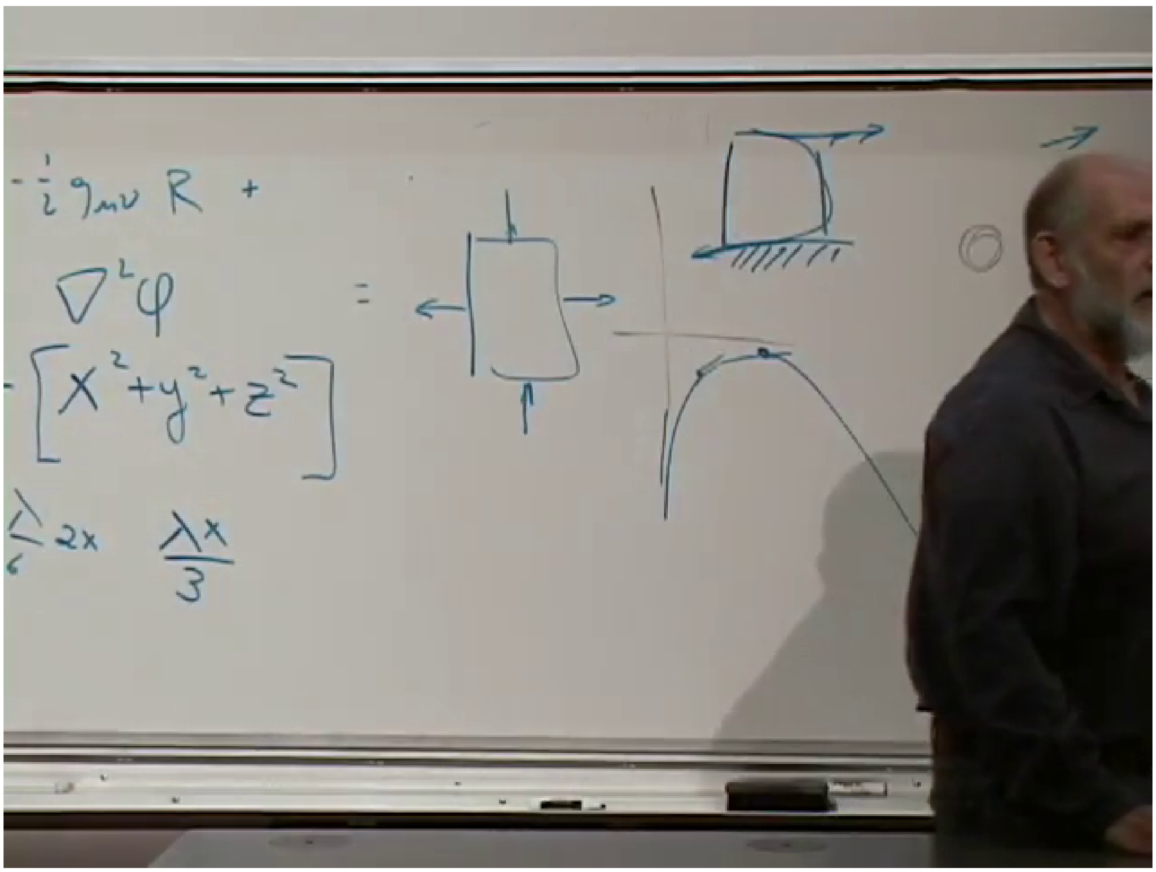

Figure 1.21: The shear square and its rotated principal-axis description. Off-diagonal stress in one basis is unequal diagonal stress in another.

Lecture frame: 01:44:45.

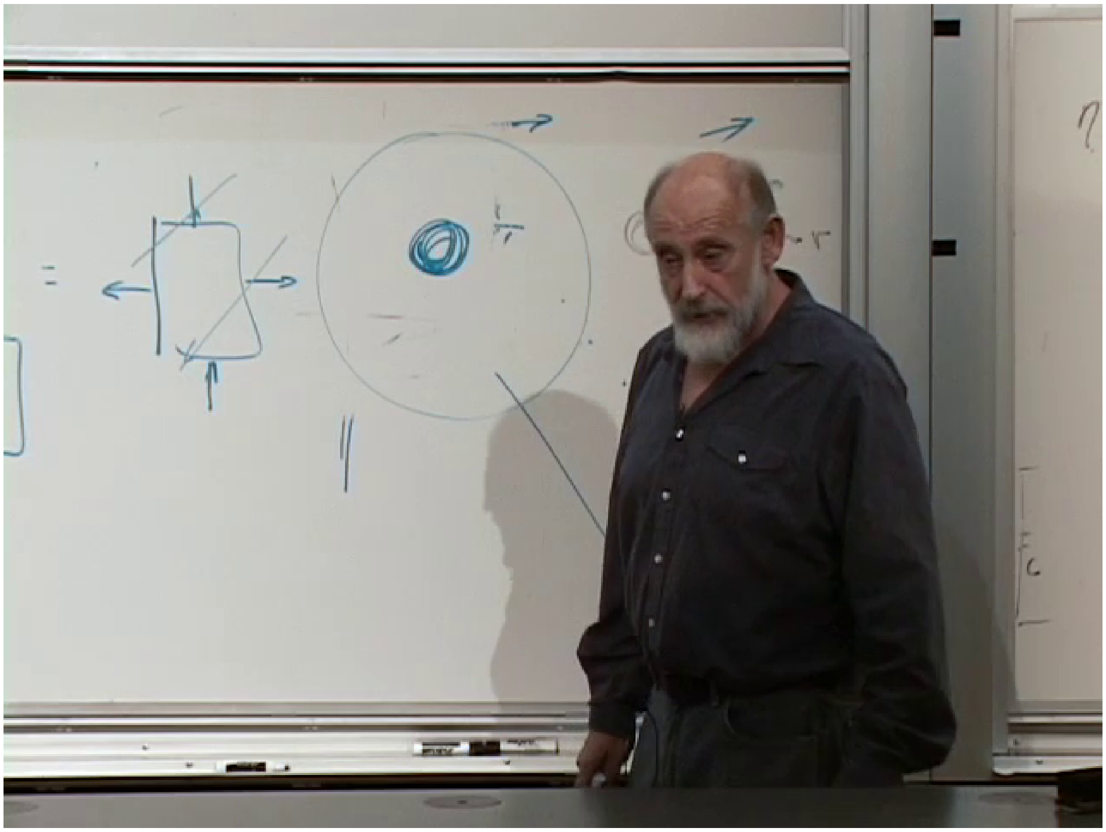

Figure 1.22: The board sketch of a circular light orbit around a compact object. For Schwarzschild geometry the photon sphere is outside the horizon at $r = 3GM/c^2$, and the orbit is unstable.

Lecture frame: 01:49:30.

Question from the audience. Can we solve intuitively for the field of a planet, and what about a genuine two-body system? [a]

Response. Spherical symmetry makes the one-body exterior manageable and leads to Schwarzschild. A strongly gravitating, time-dependent two-body system has no comparable closed elementary formula; modern calculations use numerical relativity.

[a] Lecture timestamp: 01:45:37–01:47:01.

Light can orbit a Schwarzschild black hole, but the circular null orbit is unstable. If

$$r_s = \frac{2GM}{c^2} \tag{1.41}$$

denotes the horizon radius, the photon sphere lies at

$$\boxed{r_{\mathrm{ph}} = \frac{3GM}{c^2} = \frac{3}{2}r_s.} \tag{1.42}$$

This is the corrected value previewed in the discussion: three halves of the horizon radius, not three horizon radii.

Question from the audience. Can enough light bind itself into a ring, and would an observer see photons circling a black hole? [a]

Response. At sufficiently high energy density, light's own gravity cannot be ignored, although constructing a self-consistent configuration requires the full nonlinear equations. Around a Schwarzschild black hole, null geodesics do admit the unstable circular orbit of Eq. (1.42). Detailed appearance depends on the observer and on neighboring rays, so the optical picture is deferred until the Schwarzschild geometry is in hand.

[a] Lecture timestamp: 01:47:04–01:51:50.

1.11 Eigenvalues, Vacuum Curvature, and Two Kinds of Flatness

At a regular material point, the stress-energy tensor can often be understood through its eigenvectors. In the local rest frame there is one timelike eigenvector with eigenvalue equal to the energy density. The three spacelike eigenvectors define principal stress directions. Equal spatial eigenvalues give isotropic pressure; unequal ones give anisotropic stress.

Question from the audience. What do the eigenvalues of the stress-energy tensor mean physically? [a]

Response. The timelike eigenvector identifies the comoving frame and its eigenvalue is the rest-frame energy density. The spacelike eigenvalues are principal pressures. When all three agree, the material is isotropic; when they differ, it carries anisotropic stress or shear.

[a] Lecture timestamp: 01:52:02–01:53:17.

Now consider vacuum with a cosmological constant. From

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 0, \tag{1.43}$$

contract with $g^{\mu\nu}$ in four spacetime dimensions:

$$R - \frac{1}{2}(4)R + 4\Lambda = 0,$$
$$R = 4\Lambda, \tag{1.44}$$
$$R_{\mu\nu} = \Lambda g_{\mu\nu}. \tag{1.45}$$

The sign follows the placement of $+\Lambda g_{\mu\nu}$ in Eq. (1.43); moving that term or choosing the opposite curvature convention changes intermediate signs. The invariant lesson is that nonzero Λ supports a vacuum of constant Ricci curvature.

Question from the audience. Can an isolated system be asymptotically flat when $\Lambda \neq 0$? [a]

Response. Not in the ordinary Minkowski sense. For $\Lambda = 0$, the metric of an isolated source may approach $\eta_{\mu\nu}$ as $r \to \infty$. With positive Λ, the vacuum approaches de Sitter behavior; with negative Λ, it approaches anti-de Sitter behavior. The cosmological term eventually dominates over the localized source.

[a] Lecture timestamp: 01:53:30–01:56:05.

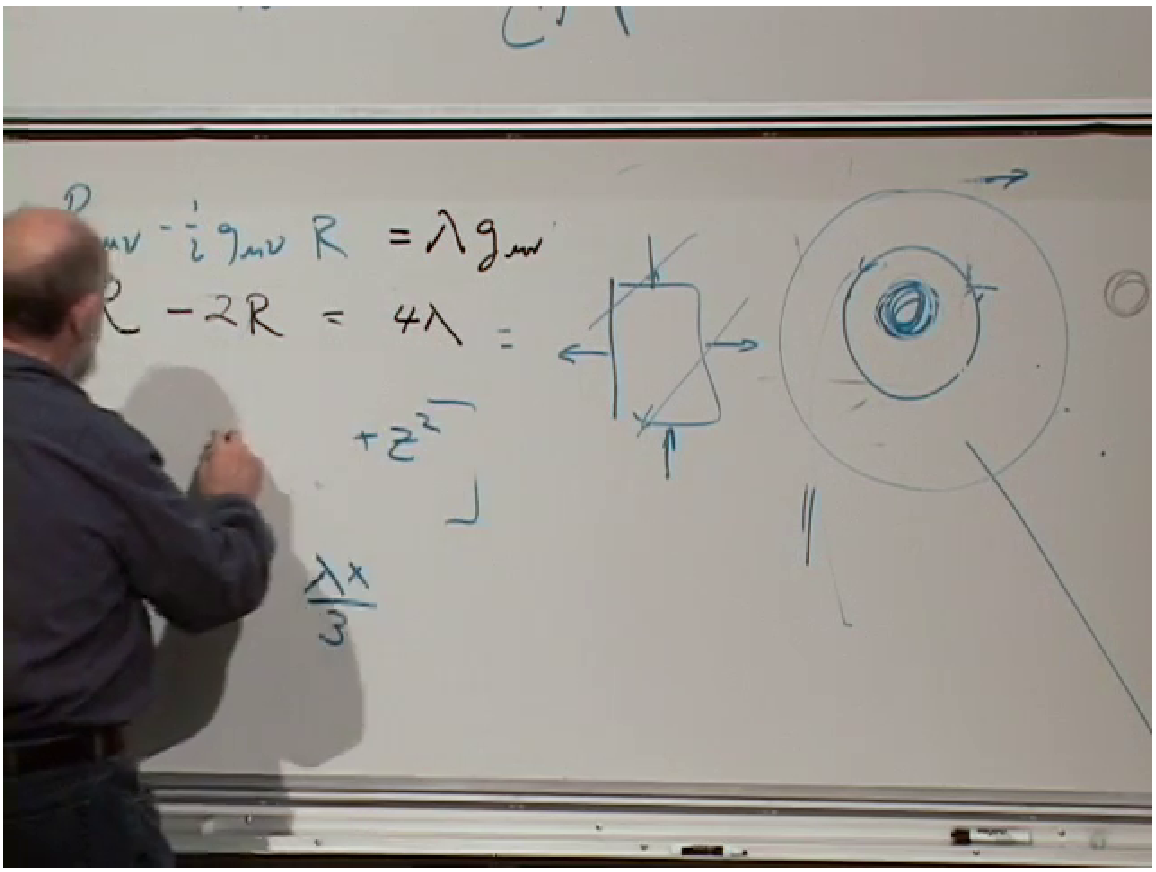

Figure 1.23: The vacuum Einstein equation with Λ and its trace on the board. With the convention used here the clean result is $R = 4\Lambda$ and $R_{\mu\nu} = \Lambda g_{\mu\nu}$.

Lecture frame: 01:55:15.

Question from the audience. Can a universe with a cosmological constant be spatially finite? [a]

Response. Yes. Closed spatial geometries are possible, but finiteness is not determined by Λ alone. Matter content, initial conditions, and the chosen cosmological solution all enter. The detailed classification belongs to cosmology.

[a] Lecture timestamp: 01:56:05–01:56:39.

Question from the audience. When we say the universe is flat, do we mean that spacetime is flat? [a]

Response. Not necessarily. Cosmologists often say that a constant-time spatial slice has nearly zero intrinsic curvature. Spacetime can still be curved because the scale factor evolves and because matter or Λ is present. By contrast, asymptotic flatness for an isolated system means that the full spacetime metric approaches the Minkowski metric far from the source.

[a] Lecture timestamp: 01:56:41–01:58:29.

1.12 What Has Been Assembled

The lecture began with a cone because it is the cleanest example of geometry remembering a source. It then enlarged that lesson in three directions.

1. Conservation selects the Einstein tensor but still permits the cosmological term:
$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$
2. The equivalence principle turns special relativity into a construction rule. For a scalar wave it yields
$$\Box_g \phi = 0.$$
3. The wave carries the conserved stress-energy
$$T_{\mu\nu} = \nabla_\mu \phi \, \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla_\alpha \phi \, \nabla^\alpha \phi,$$
so the field and geometry must be solved together.

Mass density is only the first entry in the source tensor. Momentum, pressure, shear, radiation, and vacuum energy all gravitate. Vacuum outside matter need not be flat, and a nonzero cosmological constant changes the very notion of the distant vacuum. With that structure in place, the next lecture can stop speaking abstractly and solve the first central example: the Schwarzschild geometry of a spherical mass, with cosmological expansion waiting just beyond it.